

Teaching guide: This PDF uses a small set of real items to show the main decision paths in the replenishment engine.

Model setup used: Active Demand with Active Variability, Rules Only, Summer seasonality on, and the current live service-level defaults from the dashboard.

Representative Recommendation Scenarios for the AI Team

Purpose

This document is a short teaching set. Instead of reviewing a long random item list, it uses a small number of real items that each show a different rule path in the replenishment engine.

The goal is to help programmers understand **why** the model makes a recommendation, not just **what** number it outputs.

Scope Used For This Document

- Source data: current `DataV3.xlsx` import snapshot cached by the dashboard on March 23, 2026.
- Method: `Active Demand with Active Variability`.
- Forecast mode: `Rules Only`.
- Seasonality: `On`, planning season `Summer`.
- Service levels: `A=99%`, `B=92%`, `C=88%`, `D=84%`, `E=80%`, `X=80%`.
- Low-cost local deviation threshold: `$1.00 MAC`.
- Highest-month normalization: `Off`.
- Location `1` is the DC plus a stocking location.
- Branch locations are `30`, `40`, `115`, `116`, `117`, `118`, and `119`.
- Location `9999` is ignored.

How To Read These Examples

- `Recommended Min` is the working stock the model wants to protect for that location.
- `Recommended Max` is the replenishment target for that location right now.
- Branches are usually sized for local working demand, while large deviation is pooled to the DC.

- Some outputs are not demand-forecast decisions at all. They are policy decisions such as overrides, supplier minimums, regional rules, or alert flags.

Scenario Summary

Scenario	Example item	What it teaches
Simple baseline / no special exception	ESP COMPANY \ WATERHOSE-25-PREM	What the engine does when there is no override, no regional rule, no low-cost exception, and no seasonal ramp.
Seasonal ramp to current replenishment window	COPELAND LP - WHITE RODGERS DIVISION \ BFK163S-COPF	Strong seasonal items should ramp toward the current window, not hold the full-season peak all season long.
Intermittent branch spike pooled to DC	iGAS USA, inc \ R-507-25LB-REFG	A branch should not look like a steady 16-per-month item just because it sold 16 in one isolated month.
Regional zero-max branches stay pooled to DC	TWENTYTHREEC LLC (OXB0X) \ BAYHTRJ508BRKAA-OXBX	If status contains Regional and branch current max is 0, the branch stays non-stocked.
Branch current max is zero but model wants stock	ICEO-MATIC (ICEO) \ 9101379-01-ICEO	The model can flag a branch that has no current max but now looks like it needs stocking.
Low-cost item keeps deviation local	SELECTA PRODUCTS INC \ PC2400-DURA	Cheap items under the MAC threshold keep their deviation at the branch instead of pushing it to the DC.
Manual override wins	EWCC CONTROLS, INC. \ SAS-EWCC	If Override = Y, the override location keeps its current max.
Supplier minimum amount floors the max	SELECTA PRODUCTS INC \ LN91-BATT	If S. Min Amt > 0, any positive recommended max must be at least that value.
DC gets a new max from pooled demand while current DC max is zero	NORTEK PARTS \ 01-0085-NORP	The model should flag when DC stock is being created only because branches pooled demand back to location 1.

1. Simple Baseline / No Special Exception

Item

- Supplier: ESP COMPANY
- Item: WATERHOSE-25-PREM
- Description: HOSE WATER 5/8X25FT PREMIUM

- Status: **New**
- Company totals: current min/max **8 / 16** -> recommended min/max **4 / 12**

What happened

- This item is the cleanest simple example in the current file.
- No override fired.
- No regional rule fired.
- No low-cost local rule fired.
- No supplier minimum floor fired.
- No seasonal ramp fired.
- No intermittent branch protection fired.

Important locations

- Location **1** : current min/max **1 / 2** -> recommended **4 / 5**
- Locations **30 , 40 , 115 , 116 , 117 , 118 , 119** : current max **2** -> recommended max **1**
- Branch recommended mins fell to **0**

Plain-English reasoning

- The item has very light branch demand in the scoped window.
- The branches do not justify local working stock beyond a token max.
- The DC remains the main stocking point because that is where the actual active demand signal exists.

Programmer takeaway

- This is a good reference case for the engine's default behavior when no exception rules are needed.
- It is useful as a control example when validating later changes.

2. Seasonal Ramp To The Current Replenishment Window

Item

- Supplier: **COPELAND LP - WHITE RODGERS DIVISION**
- Item: **BFK163S-COPF**
- Description: **3/8 SXS BIFLOW FILTER DRIER**
- Status: **Normal**
- Company totals: current min/max **11 / 15** -> recommended min/max **5 / 13**

What happened

- One location had enough real summer history to use the new current-window seasonal ramp.

- The item's peak month is July , but the planning window at the time of this run was the upcoming April replenishment window.
- Because of that, the branch should not hold full July-level stock yet.

Important locations

- Location 40 : current max 13 -> recommended max 4
- Location 40 : raw filtered mean 9.36 -> ramped filtered mean 3.50
- Location 40 : ramp factor 0.373786
- Location 40 : ramp window Apr-26 (28d)
- Location 1 : current max 2 -> recommended max 9 , absorbing 3.0 pooled deviation

Plain-English reasoning

- The model recognized that this is a real seasonal item with enough history to trust the monthly shape.
- Instead of using the whole summer average, it used the demand expected during the next protection window.
- That lowered the branch max now, but still allowed the DC to hold pooled uncertainty.

Programmer takeaway

- This is the clearest example of the new "current replenishment window" logic.
- The model should ramp into the season rather than acting like the peak month is already here.

3. Intermittent Branch Spike Pooled To DC

Item

- Supplier: iGAS USA, inc
- Item: R-507-25LB-REFG
- Description: AZ-50 R-507A
- Status: Normal
- Company totals: current min/max 8 / 29 -> recommended min/max 7 / 24

What happened

- This is the best real example of the intermittent-branch protection rule.
- A branch sold a large quantity in one isolated month, which would have badly overstated steady branch demand if the model simply excluded zeros.

Important locations

- Location 119 : current max 2 -> recommended max 3
- Location 119 : raw filtered mean 16.0

- Location 119 : protected filtered mean 2.67
- Location 119 : intermittent rule reason: it only sold in 1 of 6 scoped months, ADI = 6.0 , top two months = 100% of scoped demand
- Location 119 : 13.42 units were pooled back to the DC
- Location 1 : current max 21 -> recommended max 18
- Location 1 : absorbed pooled deviation 15.10

Plain-English reasoning

- Without protection, the branch would look like a steady 16 per month item because the active-demand method ignores zero months.
- The intermittent rule recognizes that this was not steady branch demand.
- It converts the branch to a much smaller inclusive mean and sends the spike stock back to the DC.

Programmer takeaway

- This is one of the most important branch-vs-DC protections in the whole engine.
- It prevents isolated spike months from turning into permanent branch stocking.

4. Regional Zero-Max Branches Stay Pooled To DC

Item

- Supplier: TWENTYTHREEC LLC (OXBOX)
- Item: BAYHTRJ508BRKAA-OXBX
- Description: HEAT STRIP 8KW JMM4 AIR HAND
- Status: Regional
- Company totals: current max 4 -> recommended max 3

What happened

- This is the cleanest current example of the Regional branch rule.
- Several branches had current max 0 .
- Because the status contains the word Regional , those zero-max branches stayed non-stocked.

Important locations

- Location 40 : current max 0 -> recommended max 0
- Location 115 : current max 0 -> recommended max 0
- Location 116 : current max 0 -> recommended max 0
- Location 117 : current max 0 -> recommended max 0
- Location 119 : current max 0 -> recommended max 0

- Those five branches were marked with `regional_zero_max_branch_pool_applied = True`
- Location 30: current max 2 -> recommended max 1, pooling 0.83 to the DC
- Location 1: current max 0 -> recommended max 1, absorbing 0.83

Plain-English reasoning

- The business meaning is: this is a regional item, not a normal branch-stock item.
- If a branch currently has no max, the model should not create new branch stock there just because demand exists somewhere in the network.
- Instead, the demand is serviced through the DC.

Programmer takeaway

- This is a policy rule, not a statistical rule.
- It should override normal branch stocking logic for statuses that contain the word `Regional`.

5. Branch Current Max Is Zero But The Model Wants Stock

Item

- Supplier: ICE0-MATIC (ICE0)
- Item: 9101379-01-ICE0
- Description: REED SWITCH
- Status: Normal
- Company totals: current max 1 -> recommended max 3

What happened

- This is the best simple alert example for a branch with current max 0 that the model now wants to stock.
- The location should be highlighted for review because it changes the stocking pattern.

Important locations

- Location 118: current min/max 0 / 0 -> recommended 1 / 2
- Location 118: `branch_zero_max_recommendation_alert = True`
- Location 1: stayed at max 1
- All other branches stayed at 0

Plain-English reasoning

- The item is not regional, so the zero-max regional protection does not apply.
- The normal demand and policy math says location 118 now needs working stock.
- Because the branch had no current max before, the model raises an alert for review.

Programmer takeaway

- This should be treated as a workflow flag, not just a number.
- The output is telling the buyer, "the engine wants to create a new stocked branch location."

6. Low-Cost Item Keeps Deviation Local

Item

- Supplier: SELECTA PRODUCTS INC
- Item: PC2400-DURA
- Description: BATTERY ALK AAA PROCELL
- Status: Bulk
- Company totals: current max 648 -> recommended max 813

What happened

- This item has MAC values around 0.51 to 0.55.
- Because the dashboard threshold was \$1.00, the low-cost local rule applied to all seven branches.
- That means the branches kept their deviation locally instead of sending it to the DC.

Important locations

- Location 30 : current max 168 -> recommended max 231
- Location 117 : current max 24 -> recommended max 71
- Location 119 : current max 24 -> recommended max 26
- All seven branches had low_cost_local_deviation_applied = True
- Pooled deviation to DC was 0 at every branch
- Location 1 : current max 144 -> recommended max 158

Plain-English reasoning

- For cheap items, it is usually acceptable to keep extra stock local because the cost of decentralizing that stock is low.
- The engine intentionally stops pooling deviation to the DC when the item cost is below the configured threshold.

Programmer takeaway

- This is another policy layer, not a demand-estimation change.
- The demand signal may be the same, but the placement of the uncertainty stock changes because of item cost.

7. Manual Override Wins

Item

- Supplier: EWC CONTROLS, INC.
- Item: SAS-EWCC
- Description: SENSOR AIR SUPPLY
- Status: Normal
- Company totals: current max 34 -> recommended max 33

What happened

- One location had a manual override.
- That location kept its current max regardless of what the normal math might have recommended.

Important locations

- Location 119 : current max 30 -> recommended max 30
- Location 119 : override_flag = True
- Location 119 : Per = HOLLADAY
- Location 1 : current max 2 -> recommended max 1
- All other locations followed the normal rules

Plain-English reasoning

- Overrides are buyer-owned decisions.
- When the file says Override = Y, the model does not try to outvote that row.
- The location stays at the current max and is marked so the UI can explain why it did not move.

Programmer takeaway

- Overrides should be treated as hard constraints unless the business explicitly decides otherwise.
- The AI model can explain them, but it should not silently ignore them.

8. Supplier Minimum Amount Floors The Max

Item

- Supplier: SELECTA PRODUCTS INC
- Item: LN91-BATT
- Description: BATTERY LITHIUM AA ENERGIZER INDUSTRIAL
- Status: New

- Company totals: current max 192 -> recommended max 96

What happened

- Every stocked location had S. Min Amt = 12.
- The demand math wanted a much smaller result, but the supplier minimum amount forced each positive recommended max to be at least 12.

Important locations

- Every location: current max 24 -> recommended max 12
- Every location: supplier_min_amount_floor_applied = True
- Every location: supplier_min_amount = 12

Plain-English reasoning

- This is not saying demand requires 12.
- It is saying the item cannot be recommended below the supplier's minimum stocking or order requirement once a positive max is justified.
- The supplier floor is acting as the last guardrail after the demand logic runs.

Programmer takeaway

- This is a post-calculation policy floor.
- It is important that the AI team not confuse this with demand itself.

9. DC Gets A New Max From Pooled Demand While Current DC Max Is Zero

Item

- Supplier: NORTEK PARTS
- Item: 01-0085-NORP
- Description: CAP 45+5MFD 370V FT4BD036
- Status: Normal
- Company totals: current max 1 -> recommended max 3

What happened

- The only meaningful branch signal came from location 115.
- That branch kept max 1, but it also pooled 1.0 unit of deviation to the DC.
- Because location 1 had current max 0, the DC had to be created by pooled demand alone.

Important locations

- Location 115 : current max 1 -> recommended max 1, pooled 1.0 to DC
- Location 1 : current max 0 -> recommended max 2
- Location 1 : dc_pooling_zero_max_alert = True

Plain-English reasoning

- The engine is saying the network needs DC coverage even though the DC is not currently stocked.
- That is a meaningful workflow event because it creates a new DC max solely from pooled uncertainty.

Programmer takeaway

- This should be highlighted, not just calculated.
- It means the system is introducing a new DC stocking decision rather than merely resizing an existing one.

What These Examples Show The AI Team

- Some outputs are mainly driven by demand estimation.
- Some outputs are mainly driven by inventory policy.
- Some outputs are mainly driven by explicit user control.
- A good AI replacement model would need to learn all three layers, or keep the policy layers as hard rules around a learned demand model.

Suggested Feature Groups For A Learned Model

- Demand history by item and location
- Seasonality by item, product group, and calendar month
- Branch vs DC role
- Lead time and supplier frequency
- ABC service level
- Status text, especially anything containing Regional
- Current min and current max
- MAC and supplier minimum amount
- Override flags and override metadata
- Intermittent-demand indicators such as active-month count, ADI, and top-two-month concentration

Suggested Training Caution

- Do not train the AI model to treat every final max as a pure forecast target.

- Many recommendation outputs are policy-adjusted targets, not raw demand expectations.
- If the AI team wants the clearest design, forecast demand first, then apply the policy rules as explicit constraints.